

James King

Portfolio: <https://jking154.github.io/JKing154/index.html>

Watertown, NY | 315-921-8281 | james.king.usa2@gmail.com | [linkedin.com/in/jamesk2](https://www.linkedin.com/in/jamesk2) | British-American dual citizen

EDUCATION

Cornell University

Ithaca, NY

Bachelor of Science Degree in Mechanical Engineering (3.4GPA)

Anticipated May 2028

Relevant Coursework: Thermodynamics | Statics | Mechanical Design | Dynamics | Waves & Oscillations | FEA |

ENGINEERING EXPERIENCE

Knorr-Bremse (New York Air Brake)

Watertown, NY

Mechanical Engineering Testing Intern

May 2026 – Present

- Designing a new iteration of an impact test bench in Ansys, analyzing the full rack and its fast-moving components for 60 mph safety requirements to ensure reliability for future product validation.
- Sourcing, redesigning, and producing drawings for a \$10K+ hose flex test rack, building two additional units for deployment to international locations to standardize testing across the company.

Cornell Racing Formula SAE (Society of Automotive Engineers)

Ithaca, NY

Ergonomics Lead / Part Designer

Nov 2024 - Present

- Lead the Ergonomics subteam, owning the design of all driver-interface systems and personally designing the head restraint, seat, and firewall to be lighter, driver-comfortable, and rules compliant for ARG27.
- Designed and fabricated the 2026 competition steering wheel through multiple driver-tested iterations, cutting weight 30% from the previous year to 330g while improving grip ergonomics.
- Ran three-point bend tests across four carbon fiber sandwich layup configurations, selecting the lightest panel that exceeded the 700 N structural threshold, achieving a 32.1 N/g strength-to-weight ratio.
- Characterized thermal conductivity of proprietary brake pad materials, delivering test data that supported the team's design event technical review and justified final material selection.

Shaw Harbor Holdings

Adams, NY

Mechanical Engineering Intern

Summer 2024 & 2025

- Designed and modeled CAD laser fixtures for manufacturing at Skip-Line LLC., reducing machine downtime by 25% and improving production efficiency.
- Co-developed enclosure prototypes in CAD that standardized components across product lines, reducing tariff exposure and cutting time-to-manufacture.

ACTIVITIES

CORE (Club of Rural Engineers)

Ithaca, NY

Member / Mentor

Sept 2024 - Present

- Mentor incoming rural students navigating college engineering programs and serve on the student board to expand resources and recruiting reach for underrepresented rural applicants.

Engineering Dean's Advisory Council (EDAC)

Ithaca, NY

Member

Sept 2025 - Present

- Selected by Admissions as one of 20 undergraduates to advise the Engineering Dean on academic policy and student experience initiatives.

ADDITIONAL

Design & Analysis: SolidWorks, Inventor, OnShape, NX | FEA (Ansys) | GD&T & Dimensional Drawings | MATLAB | Python | First Principal Analysis |

Fabrication & Testing: Carbon Fiber Composites | Test Bench Design | Machining (Mill, Lathe) | Thermal & Mechanical Testing | 3D Printing | Microcontrollers | DAQ |